

X

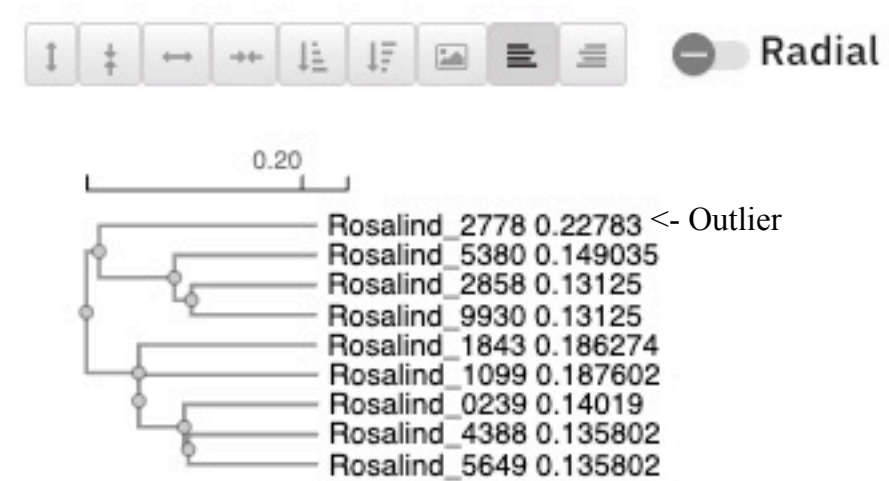
clustalo-l20260311-004012-0254-54946288-p1m

Resubmission

Guide Tree

```
(
(
Rosalind_2778:0.22783
,
(
Rosalind_5380:0.149035
,
(
Rosalind_2858:0.13125
,
Rosalind_9930:0.13125
):0.0177855
):0.0787946
):0.013284
,
(
Rosalind_1843:0.186274
,
(
Rosalind_1099:0.187602
,
(
Rosalind_0239:0.14019
,
(
Rosalind_4388:0.135802
,
Rosalind_5649:0.135802
):0.00438793
):0.0474119
):-0.00132784
):0.0548397
)
);
```

Phylogram



Selected **0** branches with current label


```
1  # Name: Christopher A. Lee
2  # Date: 03/10/2026
3  # Prof: Dr. Callura
4  # Course: Bioinformatics - ENBC 455
5
6  # Libraries
7  from Bio import SeqIO
8
9  # Problem 2
10 with open('hw6_2.txt', 'r') as f:
11     seq2 = list(SeqIO.parse(f, "fasta"))
12
13     for i in range(len(seq2)):
14         for j in range(len(seq2)):
15             if i == j:
16                 continue
17
18             if seq2[i].seq[-3:] == seq2[j].seq[:3]:
19                 print(seq2[i].id, seq2[j].id)
20     print()
21
22 # Problem 3
23 with open('hw6_3.txt', 'r') as f:
24     nodes = int(f.readline())
25     edges = 0
26     for i in f.readlines():
27         edges += 1
28
29     print(nodes - 1 - edges, nodes - 2, "\n")
30
31 # Problem 4
32 with open('hw6_4.txt', 'r') as f:
33     seq4 = [str(fasta_seq.seq) for fasta_seq in SeqIO.parse(f, "fasta")]
34
35     matrix = []
36     for i in range(len(seq4)):
37         row = []
38         for j in range(len(seq4)):
39             mismatches = 0
40
41             for k in range(len(seq4[0])):
42                 if seq4[i][k] != seq4[j][k]:
43                     mismatches += 1
44
45             row.append(mismatches / len(seq4[0]))
46
47         matrix.append(row)
48
49     for row in matrix:
50         for value in row:
51             print(f"{value:.5f} ", end="")
52     print()
```

Rosalind_9743	Rosalind_4993
Rosalind_9743	Rosalind_0997
Rosalind_6258	Rosalind_8073
Rosalind_1537	Rosalind_3718
Rosalind_1537	Rosalind_6390
Rosalind_1537	Rosalind_6399
Rosalind_1668	Rosalind_8818
Rosalind_1668	Rosalind_2715
Rosalind_1668	Rosalind_1956
Rosalind_4993	Rosalind_4416
Rosalind_4993	Rosalind_9128
Rosalind_0308	Rosalind_6644
Rosalind_5577	Rosalind_8054
Rosalind_1847	Rosalind_3718
Rosalind_1847	Rosalind_6390
Rosalind_1847	Rosalind_6399
Rosalind_1889	Rosalind_6258
Rosalind_1889	Rosalind_2499
Rosalind_1889	Rosalind_3877
Rosalind_1501	Rosalind_9891
Rosalind_1501	Rosalind_4059
Rosalind_8073	Rosalind_6888
Rosalind_0020	Rosalind_4404
Rosalind_1631	Rosalind_8818
Rosalind_1631	Rosalind_2715
Rosalind_1631	Rosalind_1956
Rosalind_2466	Rosalind_4264
Rosalind_9972	Rosalind_5627
Rosalind_9972	Rosalind_9338
Rosalind_8054	Rosalind_6258
Rosalind_8054	Rosalind_2499
Rosalind_8054	Rosalind_3877
Rosalind_5360	Rosalind_4416
Rosalind_5360	Rosalind_9128
Rosalind_9891	Rosalind_4636
Rosalind_9891	Rosalind_1658
Rosalind_5712	Rosalind_8818
Rosalind_5712	Rosalind_2715
Rosalind_5712	Rosalind_1956
Rosalind_8797	Rosalind_9796
Rosalind_2826	Rosalind_7454
Rosalind_2826	Rosalind_9972
Rosalind_2826	Rosalind_5712
Rosalind_2826	Rosalind_7636
Rosalind_6061	Rosalind_9120
Rosalind_6061	Rosalind_7710
Rosalind_1034	Rosalind_6258
Rosalind_1034	Rosalind_2499
Rosalind_1034	Rosalind_3877
Rosalind_0211	Rosalind_6061
Rosalind_0211	Rosalind_0928
Rosalind_1956	Rosalind_8561
Rosalind_1956	Rosalind_4495
Rosalind_2499	Rosalind_1501
Rosalind_2499	Rosalind_2384
Rosalind_0051	Rosalind_9120
Rosalind_0051	Rosalind_7710
Rosalind_0997	Rosalind_0308
Rosalind_0997	Rosalind_5577
Rosalind_0997	Rosalind_5312
Rosalind_4787	Rosalind_0308
Rosalind_4787	Rosalind_5577
Rosalind_4787	Rosalind_5312
Rosalind_3239	Rosalind_1248
Rosalind_3239	Rosalind_8903
Rosalind_6786	Rosalind_5360
Rosalind_6786	Rosalind_0051
Rosalind_2384	Rosalind_1248
Rosalind_2384	Rosalind_8903
Rosalind_1658	Rosalind_8818
Rosalind_1658	Rosalind_2715
Rosalind_1658	Rosalind_1956
Rosalind_6390	Rosalind_5078
Rosalind_6390	Rosalind_4665
Rosalind_6399	Rosalind_1631
Rosalind_6399	Rosalind_4787
Rosalind_6399	Rosalind_2753
Rosalind_6399	Rosalind_7823
Rosalind_6399	Rosalind_4447
Rosalind_1821	Rosalind_4993
Rosalind_1821	Rosalind_0997
Rosalind_2753	Rosalind_8818
Rosalind_2753	Rosalind_2715

Rosalind_2753 Rosalind_1956
Rosalind_7746 Rosalind_5998
Rosalind_7746 Rosalind_1821
Rosalind_7746 Rosalind_5508
Rosalind_0877 Rosalind_6888
Rosalind_8561 Rosalind_6888
Rosalind_4264 Rosalind_6644
Rosalind_4059 Rosalind_1537
Rosalind_4059 Rosalind_7398
Rosalind_4059 Rosalind_7746
Rosalind_4059 Rosalind_0798
Rosalind_6628 Rosalind_0211
Rosalind_6628 Rosalind_3239
Rosalind_9048 Rosalind_6888
Rosalind_0251 Rosalind_9891
Rosalind_0251 Rosalind_4059
Rosalind_4495 Rosalind_2466
Rosalind_4495 Rosalind_6744
Rosalind_4495 Rosalind_6896
Rosalind_1700 Rosalind_0211
Rosalind_1700 Rosalind_3239
Rosalind_8574 Rosalind_1919
Rosalind_6381 Rosalind_8818
Rosalind_6381 Rosalind_2715
Rosalind_6381 Rosalind_1956
Rosalind_3877 Rosalind_1034
Rosalind_7823 Rosalind_5078
Rosalind_7823 Rosalind_4665
Rosalind_6896 Rosalind_6644
Rosalind_5495 Rosalind_1034
Rosalind_3591 Rosalind_6888
Rosalind_8903 Rosalind_6964
Rosalind_8903 Rosalind_5495
Rosalind_8903 Rosalind_4043
Rosalind_4043 Rosalind_1631
Rosalind_4043 Rosalind_4787
Rosalind_4043 Rosalind_2753
Rosalind_4043 Rosalind_7823
Rosalind_4043 Rosalind_4447
Rosalind_9796 Rosalind_1361
Rosalind_9796 Rosalind_9048
Rosalind_0928 Rosalind_1501
Rosalind_0928 Rosalind_2384
Rosalind_5312 Rosalind_5078
Rosalind_5312 Rosalind_4665
Rosalind_4665 Rosalind_6964
Rosalind_4665 Rosalind_5495
Rosalind_4665 Rosalind_4043
Rosalind_4410 Rosalind_1537
Rosalind_4410 Rosalind_7398
Rosalind_4410 Rosalind_7746
Rosalind_4410 Rosalind_0798
Rosalind_7710 Rosalind_6786
Rosalind_4447 Rosalind_8073
Rosalind_6644 Rosalind_3549
Rosalind_6644 Rosalind_6064
Rosalind_6644 Rosalind_2138
Rosalind_3324 Rosalind_2826
Rosalind_3324 Rosalind_0877
Rosalind_9128 Rosalind_4993
Rosalind_9128 Rosalind_0997
Rosalind_2138 Rosalind_1889
Rosalind_2138 Rosalind_9382

34 924

0.00000	0.48019	0.51621	0.34214	0.49940	0.30372	0.54382	0.46339
0.48019	0.00000	0.63986	0.59664	0.63505	0.32533	0.55462	0.48019
0.51621	0.63986	0.00000	0.31573	0.47419	0.57743	0.64106	0.60744
0.34214	0.59664	0.31573	0.00000	0.34334	0.48739	0.61345	0.56903
0.49940	0.63505	0.47419	0.34334	0.00000	0.57263	0.64226	0.61465
0.30372	0.32533	0.57743	0.48739	0.57263	0.00000	0.46819	0.30732
0.54382	0.55462	0.64106	0.61345	0.64226	0.46819	0.00000	0.32413
0.46339	0.48019	0.60744	0.56903	0.61465	0.30732	0.32413	0.00000